YuE2-VAE-RS ×3

A grown decoder for the YuE2 music VAE, trained against the 100 Hz ripple and the 6 / 12 kHz seams of its Standard decoder

Viktor Zhuromskyy Ruach Studio **Claude** Anthropic

weights, study and integration guide: huggingface.co/goldhub/YuE2-RS-Finetunes · code: github.com/igrbible/Ruach_Studio

Abstract

YuE2-VAE-RS ×3 is a larger decoder for the waveform VAE of m-a-p's YuE2 music model. It reads the same latents (64 channels at 25 frames per second) and writes the same 48 kHz stereo; the encoder is m-a-p's, frozen and never run, so nothing upstream of the decoder changes. Measured against originals, the released decoder, which we call Standard, carries two traces of its upsampling: a 100 Hz ripple in the envelope above 3 kHz on sustained dense material (up to +7.2 dB over its neighbourhood, where the originals sit at +0.2 to +0.6 dB) and spectral lines at 6 and 12 kHz (up to +20.6 and +30.0 dB over the local floor). We grow the decoder from 66.4 M to 200.8 M parameters with residual units of Standard's own form whose output starts at zero, so that at step 0 it decodes as Standard to the sample, and train the whole decoder adversarially on music with losses aimed at both traces, in two runs of about three and a half days on one RTX 3090 each. Two checkpoints from the 35th and 36th hour are released, in float32 only. Over seven music clips the mean excess of the ripple falls from +2.71 dB to +0.60 / +0.36 dB, of the 6 kHz line from +3.59 to +0.98 / +0.79 dB, of the 12 kHz line from +6.96 to +2.16 / +2.13 dB, and the loss of energy at 16–20 kHz from −3.19 to −1.58 / −1.64 dB. The costs are reported as plainly: SI-SDR is lower on every clip (7.85 / 7.88 against 8.54 dB), the floor between harmonics is not better, on 40 generated songs the seams did not vanish but became a taller, narrower line inside a deeper notch, and decoding takes twice as long. We count every hour of audio and of GPU time, describe a drift of both runs after their best day that coincides with a widening ratio of the discriminators' learning rate to the generator's, and show the decoder numerically identical across PyTorch and a ggml engine on CUDA, Intel Arc and CPU (−108 to −118 dB from the reference).

> «трое или четверо суток возни с VAE не должны исчезнуть молча» three or four days of work on the VAE should not vanish without a word
>
> «мы ни от кого ничего не скрываем, но прозрачность чище хрусталя» we hide nothing from anyone, and the transparency should be clearer than crystal
>
> Viktor Zhuromskyy, on these days of work

1 INTRODUCTION

YuE2 [1] generates a song as acoustic latents, 64 numbers 25 times a second, and its VAE [2] turns them into 48 kHz stereo; its predecessor YuE [3] and the semantic supervision YuE2 draws from MERT2, a successor of MERT [4], stay upstream and are not touched here. The decoder follows the Oobleck layout of stable-audio-tools [5, 6] with the SnakeBeta activation of BigVGAN [7, 8]. Every song the model sings passes through it, so its traces are in every song. m-a-p ship two decoders for the same latents, YuE2-Vae and YuE2-Vae-legacy; we call them Standard and Legacy, and their per-tensor merge $0.666 \cdot \text{Standard} + 0.334 \cdot \text{Legacy}$ we call Mix 0.666.

Upsampling by transposed convolutions is known to leave periodic traces: checkerboard patterns in images [9] and, in neural audio, tonal artifacts at multiples of the upsampling grid [10]. Standard is good enough that, to our knowledge, nobody had fine-tuned it, yet measured against originals it carries two such traces, a ripple at four times the latent frame rate and lines at a quarter and a half of the Nyquist frequency (§2).

We fine-tune the decoder alone. The latent space stays the one YuE2 sings in: the encoder is Standard's, frozen, and the grown decoder starts as an exact copy of Standard. The contributions are:

- a measurement of the 100 Hz ripple and the 6 / 12 kHz seams in Standard, Legacy and Mix 0.666 on eight clips no training saw (§2);
- a function-preserving grown decoder, ×3 at 200.8 M parameters, and its recipe of losses, phases and checkpoint selection (§3), with every hour of audio and of GPU time counted, the training curves and the drift after the best day (§4);
- results against the originals and on 40 generated songs, including where the decoder is worse (§§ 5 and 6);
- numerical identity across PyTorch and a ggml engine, and a float32-only release under CC BY-NC 4.0 with integration code under Apache-2.0 (§§ 7 and 8).

The numbers that favour the decoder and those that do not are given side by side, each with where and how it was measured. Clip numbers are on 30-second clips decoded whole; numbers on whole songs come from decodes with the exact tiling halo of §7.3. What the differences sound like is left to the listener.

2 THE PROBLEM IN STANDARD

The YuE2 VAE turns its latents into audio through six transposed-convolution upsampling blocks, 1 920 samples per latent frame. Two traces of that structure are measurable in Standard's output. Both were measured on eight 30-second clips that no training saw, decoded whole in float32 with TF32 off and compared with their originals: 06, a shofar; 07 and 08, Estas Tonne's guitar in the studio and live; 37, a woman reading (a 24 kHz source with nothing above 12 kHz, left out of every mean); 05, a duduk; 12, Shchedrin's orchestra; 26, a Scorpions rock ballad; 34, a Petra drum solo. These are the clips of an earlier measurement of the YuE2 VAE's ceiling, which over 22 pieces found Standard losing 1.54 dB at 12–16 kHz and 2.55 dB at 16–20 kHz on average.

2.1 THE 100 HZ RIPPLE

The envelope of everything above 3 kHz, taken with a 1 ms window and divided by its own 100 ms trend, shows a peak at 100 Hz in its Welch spectrum [18]: four times the latent frame rate, the frame grid heard as a fine crackle on sustained tones. The measure is that peak over its ±3–15 Hz neighbourhood. It is not where it was first looked for, the shofar's decay (−2.2 dB, the original −1.5), but in sustained dense material:

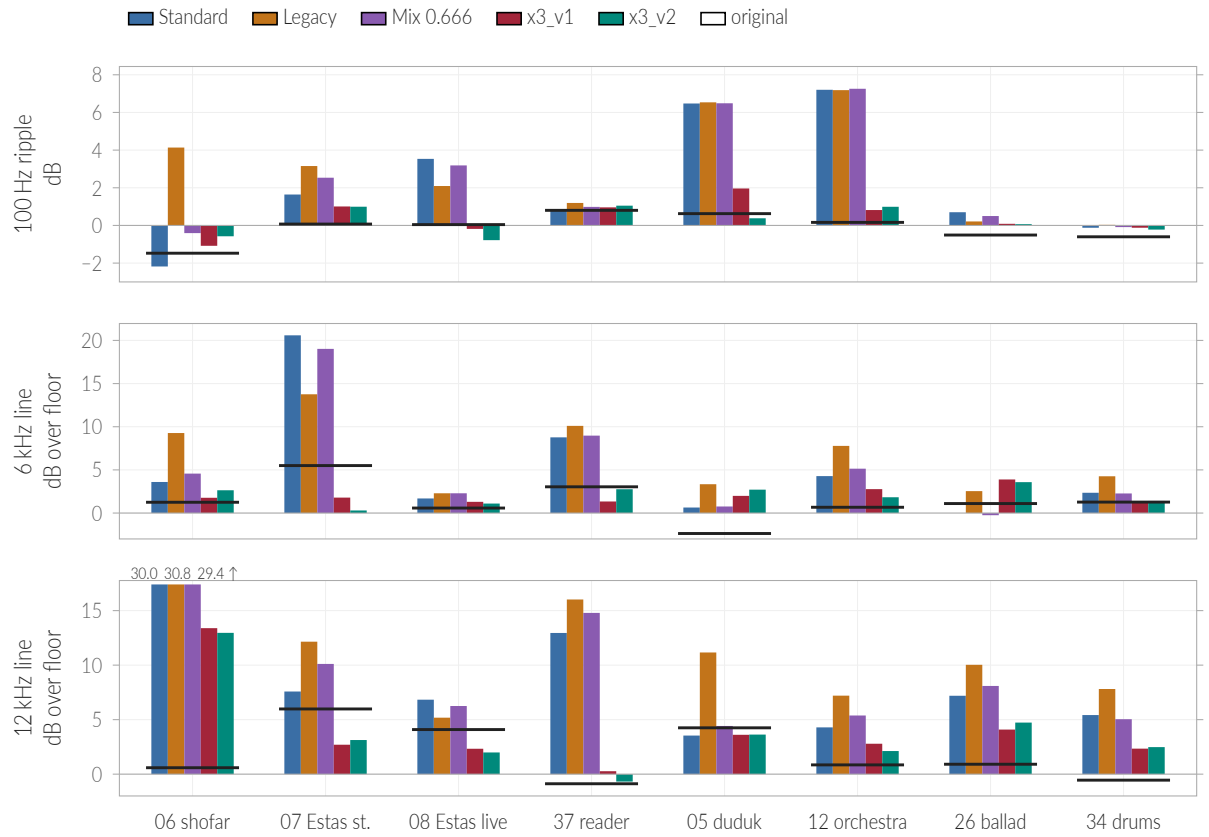


FIGURE 1 The three traces per clip, all five decoders; the black dash is the original's own value. Top: the 100 Hz ripple of the >3 kHz envelope, dB of the 100 Hz peak over its ±3–15 Hz neighbourhood (Welch); the shofar is measured over its decay only. Middle and bottom: the 6 and 12 kHz lines over their local floor. The shofar's three m-a-p bars at 12 kHz run off the top; their values are written over them. Standard, Legacy and Mix 0.666 are discussed in §2, x3_v1 and x3_v2 in §5.

the duduk at +6.5 dB, the orchestra at +7.2, Estas Tonne live at +3.5 and in the studio at +1.6, against originals at 0.0 to +0.6 (Figure 1, top). Legacy and the merge carry the same ripple on the duduk and the orchestra (+6.5 / +7.2 and +6.5 / +7.3 dB), so it is not a quirk of one checkpoint, and no merge of the two removes it.

2.2 THE SEAMS AT 6 AND 12 KHZ

6 kHz is a quarter and 12 kHz a half of the 24 kHz Nyquist frequency; lines there are traces of how the bands are stitched in the upsampling. The measure is the line (±6 Hz, Welch with 16 384-point segments) over the median of its neighbourhood 60–400 Hz away. Standard's 6 kHz line is loudest on Estas Tonne's studio recording, +20.6 dB where the original has +5.5 of its own, and +8.8 dB on the reader (original +3.0). The 12 kHz line is its largest: +30.0 dB on the shofar's decay (original +0.6), though at −109.3 dBFS; +12.9 on the reader, +7.2 on the rock ballad, +5.4 on the drums. Legacy and the merge are of the same order or worse (Figure 1, middle and bottom).

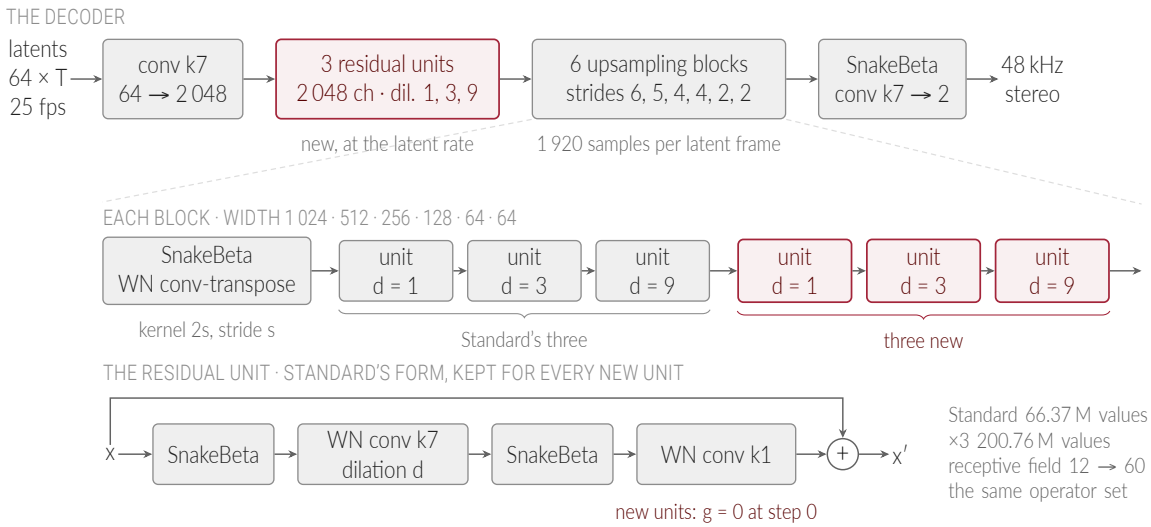


FIGURE 2 The grown decoder. Grey: Standard's path, unchanged. Red: the new residual units, three at the latent rate and three after Standard's own three in each of the six upsampling blocks. A new unit's last convolution starts with its weight-norm magnitude g at zero, so the grown decoder decodes as Standard until it learns.

3 METHOD

3.1 THE GROWN DECODER

Standard's decoder is m-a-p's Oobleck layout: a first convolution (kernel 7) from the 64 latent channels to 2 048, six upsampling blocks with strides 6, 5, 4, 4, 2, 2 and widths 1 024, 512, 256, 128, 64, 64, each a SnakeBeta, a weight-normalised [11] transposed convolution and three residual units with dilations 1, 3, 9, and a last SnakeBeta and convolution to two channels. ×3 adds three residual units at the latent rate (2 048 channels, dilations 1, 3, 9) between the first convolution and the first block, and three more units in each of the six blocks after Standard's three, at the block's own width (Figure 2). Every new unit has Standard's own form, SnakeBeta, a weight-normalised dilated convolution with kernel 7, SnakeBeta, a weight-normalised convolution with kernel 1, added to the residual stream, so the operator set is unchanged and any engine that runs Standard's kernels runs ×3.

The last convolution of each new unit starts with a weight-norm magnitude of zero, so at step 0 every new unit adds exactly zero: the gate before training found the grown decoder equal to Standard to the sample ($\max|\Delta| = 0.0$ on clips 06, 07 and 08, threshold 10^{-5}). The file holds 200.76 M stored values against Standard's 66.37 M.[1] The new units widen the decoder's receptive field from 12 to 60 latent frames on each side, which matters for tiled decoding (§7.3). A larger ×4 variant was trained beside ×3 for one day and stopped; it is not part of this release, and its hours are not counted here.

3.2 DATA: A LATENT CACHE AND A POOL

The encoder was never run. Training pairs are latents from the cache of Ruach Studio's RS-TW dataset, made earlier by the Standard encoder, and the audio each was made from, read as the cache read it (ffmpeg from the file's start, 48 kHz, no normalisation). The pairing was checked on 26 tracks from every source: median SI-SDR 7.7 dB, the codec's own ceiling rather than a shift (a one-frame shift would give about zero), and a best-correlation lag of 0 on 23 tracks and of ±1 sample on 3.

[1] All stored values of the file: 427 tensors, 200 756 034 numbers, the weights, biases, weight-norm magnitudes and SnakeBeta parameters. A count of 200 631 168 leaves out the 42 946 weight-norm magnitudes and the 81 920 SnakeBeta parameters.

TABLE 1 Reconstruction terms and their weights in the two branches.

term	what it measures	x3	x3-hf1
long	log-STFT 2 048 / 4 096 / 8 192, floor 100 dB under each frame's peak (a quiet frame weighs as a loud one)	2	2
valley	10th, 20th and 35th percentile of each frame's log power in 0.5–10 kHz: the floor between harmonics	2	2
grid	log power at the multiples of 3 kHz (8 192-STFT bins $512 \cdot k$): the upsampling grid	1	1
short	log-STFT, short windows 128–1 024	0.5	0.5
bands	1/3-octave energy, ×3 on 2–12 kHz	1	1
subbass	20–40 and 40–70 Hz	0.5	0.5
env	log peak envelopes at 5 and 10 ms	1	1
hf	the >3 kHz envelope at 1 ms: the 100 Hz ripple	0.5 → 2	1
mod	envelope modulation locked to the latent frame grid (DFT at 25 / 50 / 75 / 100 Hz from the frame edge, five bands 3–24 kHz, phasors averaged over the batch, output against original)	0 → 1	1
side, width	side signal and S/M per 1/3 octave	0.5, 0.5	0.5, 0.5
lf	complex STFT below 1.5 kHz, where the decoder copies the wave	0.3	0.3

The list holds 83 606 tracks, 5 082.8 h; its two largest sources are FMA [16] (3 143.9 h) and MTG-Jamendo [17] (1 208.1 h). Removed before training: Ruach Studio's own finished takes, which are already Standard decodes and would teach Standard's errors back; two shofar sources that might contain the held-out shofar clip; the four tracks the evaluation clips come from; and broken rows and rows without their audio. No private voice recordings and no renders of licensed virtual instruments are in it, checked by path. 48 tracks, up to four per source, are held out for validation.

A CPU process keeps a pool: a ring of 48 slots × 1 024 windows of 2.56 s (64 latent frames and a 6-frame halo on each side), 35.0 h at any moment, refilled slot by slot; 1 399 slots were written over the 3.6 days of training. Tracks are drawn per source with a weight proportional to the square root of its hours, generated songs at half weight; near-digital silence (below −75 dBFS) is skipped and quiet passages are kept. The trainer takes a slot at random, then a window, then a random crop of 32 latent frames (1.28 s) with its halo.

3.3 LOSSES

The reconstruction part is a weighted sum of the terms of Table 1. Two branches were trained. x3 changed its weights once, at step 34 547, when the weight of the 1 ms envelope above 3 kHz rose from 0.5 to 2 and the frame-grid modulation term was switched on. x3-hf1 branched from x3 at step 130 414 with that envelope's weight at 1 instead of 2, because x3 had measured brighter in the air band (8–14 kHz +0.6 to +1.7 dB over Standard) and the raised weight was the suspect.

ADVERSARIAL PART A multi-period discriminator with periods 2, 3, 5, 7, 11 [12] and a multi-scale STFT discriminator with windows 4 096 down to 256 [13], both trained from scratch, with the least-squares objective [14] and relative feature matching [15]. The generator's loss is $2 \cdot$ reconstruction + $1 \cdot$ adversarial + $2 \cdot$ feature matching. The discriminators compute in bf16; the decoder computes in float32 with TF32 off. The gradient norm is clipped at 5, and every step was clipped: the median norm before clipping was 155–173 in the windows of §4.4.

PHASES AND RATES Warm-up, steps 0–3 000: only the new units learn (rate $2 \cdot 10^{-4}$), from the reconstruction alone; the discriminators learn beside them without reaching any weight. Fade-in, 3 000–9 000:

TABLE 2 The judge's terms for Standard and the two releases, recomputed by the bench of §5. Bold: the lowest in each column (lower is better).

decoder	score	floor	midhigh	tilt	peaks	crest	line6k	ripple	subbass	width
Standard	1.952	2.86	2.27	0.32	**1.74**	3.22	3.92	**0.00**	2.95	0.29
x3_v1	**1.320**	2.94	1.35	**0.13**	1.87	**2.39**	**0.00**	0.39	**2.70**	0.11
x3_v2	1.518	**2.79**	**1.33**	0.64	1.85	3.09	**0.00**	0.90	2.97	**0.08**

every weight learns (Standard's own at $2 \cdot 10^{-5}$, the new units at 10^{-4}), and the adversarial and feature-matching terms fade in linearly. Full training from 9 000: the generator's rates halve every 100 000 steps, continuously, as $0.5^{(s\,-\,3\,000)/100\,000}$; the discriminators' rate stays at 10^{-4}. Batch 4 × 32 frames, then 8 × 32 (x3 from step 139 560, x3-hf1 from 132 388). An EMA of the generator with decay 0.999, about 21 minutes at batch 8; every evaluation and every released file is EMA weights.

GUARDS A step with a non-finite loss or norm is skipped; ten such in a hundred steps, or a reconstruction loss held above 1.8 times its median for 200 steps, stops the trainer without saving, so the last checkpoint (written every 30 minutes) stays the last good state. A restart resumes from it and the steps past it are discarded; they are counted in §4.1.

3.4 THE JUDGE AND THE CHOICE OF CHECKPOINTS

Every hour the EMA weights decode three held-out clips (06 shofar, 07 and 08 Estas Tonne) and a judge scores them against the originals: the plain mean of nine terms, each 0 when the decode matches and growing with the excess. *floor*: the shofar's floor between harmonics over the thirds of its decay; *midhigh*: band deficits of 07 and 08 in the middle-high range; *tilt*; *peaks*: the peak columns p10 / p90; *crest*; *line6k*: the 6 kHz line's excess on 07 in dBFS; *ripple*: the 100 Hz ripple's excess on the shofar's decay; *subbass*: 20–60 Hz on 07; *width*. Each evaluation also computes the reconstruction loss on held-out windows of the 48 validation tracks, with the original weights, so that it stays comparable across the weight change. Table 2 gives the terms for Standard and the two releases, recomputed on the bench of §5; they equal the training's own records to the fourth digit.

A caution about the judge, found on the way: its line6k term counts only an excess at 6 kHz. At v1 the 6 and 12 kHz lines on 07 and 08 are holes, 3–7 dB under the original (07: −6.87 dB at 6 kHz, −7.20 at 12 kHz), and the term reads 0. The seam is still there; it changed sign. §5.4 measures it finely.

Checkpoints were chosen by the judge's steadiness, not by its single best hour. A stable window is a run of five consecutive evaluations (four hours), ranked by the mean plus the standard deviation of the score; the EMA was saved every 8 hours, so a release could only be one of those snapshots (§4.3).

3.5 HARDWARE

One NVIDIA RTX 3090 (24 GB) per branch, both in one machine, PyTorch 2.10 / 2.11 [22], CUDA 12.8. A step took about 0.69 s at batch 4 and 1.25–1.27 s at batch 8. The cards were not exclusive at all times: other jobs shared the machine, which is why decode speeds in §6 are given as best and median.

4 TRAINING: HOURS, CURVES AND DRIFT

4.1 WHAT EACH CHECKPOINT SAW

The counts come from the runs' own records, not from a plan: the steps from the training log (a record every 50 steps); the batch from each record's memory (batch 4 holds 11.8–14 GB, batch 8 holds 20.9 GB); a

TABLE 3 What the two releases were trained on, and the two runs to their planned end. x3_v2 is a branch of x3: its first 185.5 h of audio and 26.2 GPU-hours are x3's up to step 130 414. The whole of both runs: 138.2 GPU-hours on RTX 3090 cards and 1 073.2 hours of audio crops.

	last step	1.28 s crops	hours of crops	of which own	GPU-hours	steps discarded	pool written, h
x3_v1 (x3, saved 7 Nov 09:03)	161 953	737 424	262.2	262.2	36.1	400	456.6
x3_v2 (x3-hf1, 7 Nov 08:00)	155 030	710 896	252.8	67.3	34.6	400	443.5
x3, whole run (83.1 h wall)	291 100	–	629.5	629.5	82.4	1 762	–
x3-hf1, whole run (56.4 h wall)	287 350	–	629.1	443.7	55.8	1 446	–

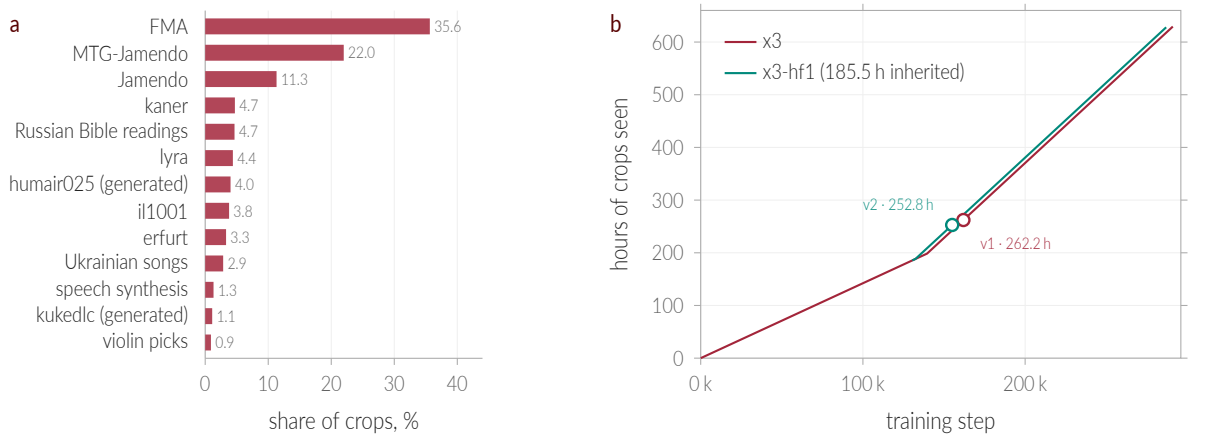


FIGURE 3 (a) Where x3_v1's 262.2 hours of crops came from, by source (x3_v2: the same shares within 0.1 %). Weighting by the square root of hours is why small sources weigh more than their size. (b) Cumulative hours of 1.28 s crops against the training step; the slope doubles where the batch went from 4 to 8. Circles: the two released checkpoints.

crop is 32 latent frames, 1.28 s of stereo audio per batch item (the halo is context, not rendered); the share of each source is the pool's per-slot mean at the moment the trainer drew from it. A restart drops the steps past the last checkpoint; those are counted apart as discarded, since no checkpoint learned from them.

The pool was refilled faster than the trainer consumed it, so the decoder almost never saw a window twice: by the time v1 was saved, 456.6 h of windows had been written, about 413.2 h of it distinct audio, and each written second was drawn 0.57 times on average. The pool's universe was 5 081.4 usable hours; the decoder saw a small, nearly unrepeated sample of it (Table 3 and Figure 3). Generated songs are 5.1 % of what v1 saw. The discarded steps: x3's first start (400 steps, restarted from zero when the valley term was added), a reboot of the machine on 7 November (103 steps in x3, 206 in x3-hf1) and a restart the same afternoon (1 259 and 1 240).

4.2 TRAINING CURVES

Figure 4 shows every hourly evaluation of both branches. The score is noisy from hour to hour, with a standard deviation of 0.04–0.12 even inside the steadiest windows. It scattered widely through the first day, while the weights and the loss terms changed, was low and steadiest on 7 November, and rose again afterwards in both branches while the held-out reconstruction kept improving (§4.4). At the planned end x3 scored 2.067 and x3-hf1 1.579, against Standard's 1.952; 20 of x3's 86 evaluations after step 0 scored worse than Standard.



FIGURE 4 Training curves of x3 (red) and x3-hf1 (teal, branched from x3 at step 130 414). From the top: the judge score, the 100 Hz ripple on the shofar's decay, and the held-out reconstruction loss (from step 2 441, after the grid term joined it). Shaded: the steadiest window of x3 and the two steadiest of x3-hf1. Dotted: end of fade-in (9 000), x3's weight change (34 547) and its batch change (139 560). Circles: the released checkpoints.

TABLE 4 The three steadiest windows of each branch, by the mean plus the standard deviation of the score over five evaluations. Bold: the lowest mean score, deviation and ripple.

branch	rank	steps	time	mean score	sd	mean ripple, dB
x3	1	150 723 – 161 954	7 Nov 05:03 – 09:03	**1.374**	0.066	+0.82
x3	2	54 392 – 74 725	6 Nov 08:10 – 12:10	1.381	0.123	+0.45
x3	3	184 302 – 195 798	7 Nov 18:06 – 22:06	1.465	0.085	+1.27
x3-hf1	1	169 225 – 180 530	7 Nov 13:01 – 18:07	1.425	**0.035**	**+0.08**
x3-hf1	2	155 032 – 166 420	7 Nov 08:01 – 12:01	1.439	0.061	+0.43
x3-hf1	3	132 388 – 143 651	6 Nov 23:59 – 7 Nov 04:01	1.473	0.080	+1.24

4.3 HOW THE TWO CHECKPOINTS WERE CHOSEN

v1 is x3's EMA snapshot at step 161 953, saved on 7 November at 09:03, at the end of x3's steadiest window; the evaluation 17 seconds later scored 1.3204 with a ripple of +0.39 dB. No snapshot of x3-hf1 fell inside its steadiest window (steps 169 225–180 530; snapshots at 132 388, 155 030, 200 359, 222 959, 245 462, 267 994 and 287 379), so v2 is the snapshot at the start of its second window, step 155 030; it scored 1.5181 with a ripple of +0.90 dB. The single lowest hour of the whole x3 run (step 54 392, score 1.210) lies in a window whose scores scattered with a standard deviation of 0.123, against 0.066 in v1's (Table 4).

TABLE 5 Medians over evaluations (judge, held-out loss) and over the log's 150-step bins (losses) in the steady stretch and in the drift window. The D ÷ G ratio follows the schedule of §3.3; for the drift window, its range in brackets. Bold, for the judge's score and terms and the two reconstruction losses: the better of the two windows in each branch; the adversarial losses, the gradient norm and the ratio have no better side.

	x3		x3-hf1	
	150–162 k	≥ 200 k	150–162 k	≥ 200 k
judge score	**1.411**	1.820	**1.493**	1.731
100 Hz ripple, dB	**1.00**	4.04	**1.08**	2.84
crest term	**2.54**	3.08	2.94	**2.87**
subbass term	**2.79**	3.10	**2.93**	3.07
width term	**0.07**	0.17	**0.06**	0.13
tilt term	**0.14**	0.31	0.58	**0.43**
floor term	2.85	**2.73**	**2.66**	3.02
midhigh term	1.24	**1.20**	**1.29**	1.41
held-out reconstruction	5.132	**5.096**	5.134	**5.097**
training reconstruction (run weights)	6.209	**6.159**	5.852	**5.806**
D's loss d	0.291	0.274	0.300	0.283
G's adversarial loss	0.553	0.584	0.540	0.568
feature matching	0.768	0.805	0.767	0.802
gradient norm before clipping	172	155	173	160
D rate ÷ G rate (grown units)	2.9	5.4 (3.9–7.4)	2.9	5.3 (3.9–7.2)

4.4 THE DRIFT AFTER THE BEST DAY

After 7 November both branches drifted: the judge's score worsened while the reconstruction kept improving on held-out windows to the last step. Table 5 compares the medians of the steady stretch (steps 150 000–162 000, where v1 was saved) with those of everything from step 200 000 to the end. Not every term moved: in x3 the floor and midhigh terms held or improved; in x3-hf1 the crest and tilt terms did. The ripple drifted first: in x3 it never came back under 2.4 dB after step 195 798, in x3-hf1 after 183 354; then the sub-bass, crest and width terms in x3 and the shofar's floor in x3-hf1. These are what no reconstruction term weighs on the judge's own pieces. The reconstruction did not drift: the held-out loss went from 5.132 to 5.096 in x3 and from 5.134 to 5.097 in x3-hf1.

WHAT CHANGED BESIDE IT The trainer halved the generator's learning rate every 100 000 steps but never the discriminators'. The ratio of the discriminators' rate to the generator's (grown units) was 2.9 in the steady window; in the drift window its median was 5.4 (3.9–7.4), and at the last step 7.4 in x3 and 7.2 in x3-hf1. These come from the schedule and agree with the optimizers' own state in x3's last checkpoint (step 291 100): the discriminators at $1.0 \cdot 10^{-4}$, the grown units at $1.36 \cdot 10^{-5}$. Against Standard's own weights, whose base rate is a fifth of the grown units', every ratio here is five times larger. With it, D's own loss fell (0.291 → 0.274 in x3: it told real from decoded better), G's adversarial loss rose (0.553 → 0.584) and the feature distance grew (0.768 → 0.805); the same in x3-hf1 (Figure 5a).

IS IT THE DATA? One hypothesis was that the swings come from what lands in the batches. Both trainers drew from one pool at the same time, so if the data drove the hourly jumps, the two branches would jump together. They do not: over 56 common hours and 55 paired hour-to-hour changes, the correlation of the two branches' score changes is r = +0.02 (Figure 5b); of the 19 hours where either jumped by more than 0.15, both moved the same way by more than 0.08 in 3. A second test asked whether a source's share in

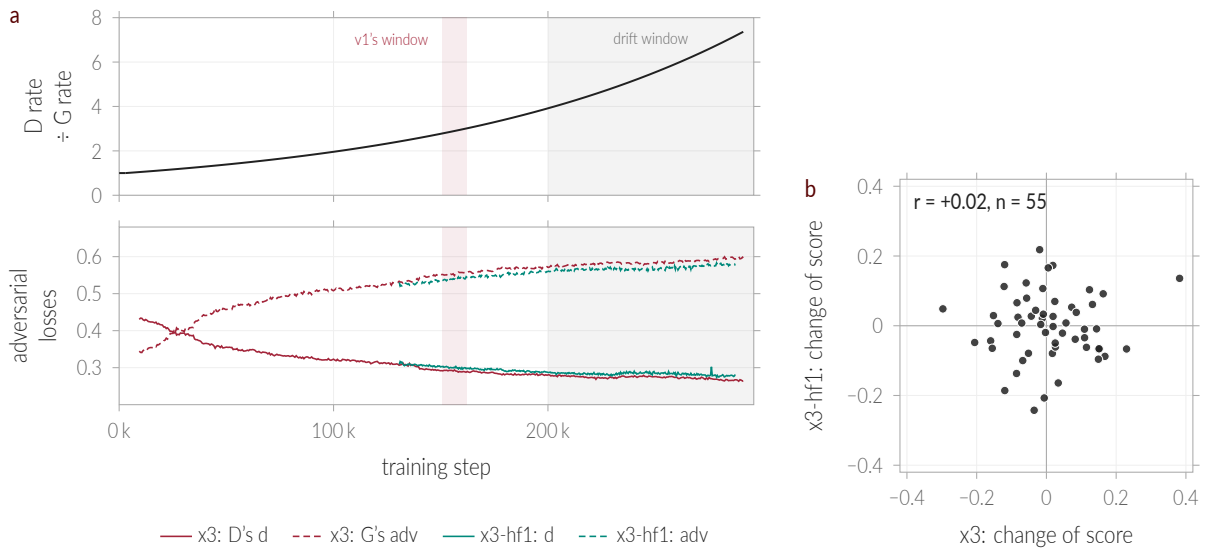


FIGURE 5 (a) Top: the ratio of the discriminators' learning rate to the generator's (grown units), from the schedule. Bottom: D's loss d (solid) and G's adversarial loss (dashed), means of 600 steps, from step 9 000. Shaded: x3's steadiest window and the drift window. (b) Hour-to-hour change of the judge score, x3 against x3-hf1, for every pair of consecutive common hours (gaps up to 3 h).

an hour's refills moves the next score: the strongest correlations were −0.29, −0.18 and −0.15 in x3 (n = 83) and +0.27, −0.23 and +0.18 in x3-hf1 (n = 55), for different sources with different signs in the two branches, none above 0.3 in size.

A CORRELATION, NOT A CAUSE The drift and the widening D : G ratio come together, twice, on one pool, with uncorrelated hourly swings. That points at each GAN's own state rather than at the data, but it does not prove the ratio is the cause: two causes can leave one trace. A generator slowing against a discriminator still at full speed is one; the decoder over-fitting the judge's blind spots after 150 000 steps is another. Telling them apart needs a run that holds the ratio and changes nothing else; this report has none, so the ratio stays a suspect, not a finding.

5 RESULTS

5.1 THE BENCH

Five decoders read the same latents of the Standard encoder: Standard, Legacy, Mix 0.666, x3_v1 and x3_v2. All are loaded into one module (the m-a-p checkpoints through a loader that fails on any key it cannot place, the ×3 files strictly) and decoded in float32 with TF32 off. The eight clips fit one tile and are decoded whole. The measures are those of the earlier ceiling study and of the judge, unchanged: band energy against the original in 0–4, 4–8, 8–12, 12–16 and 16–20 kHz; SI-SDR [19]; mel and multi-STFT log distances; the ripple and the lines of §2; the floor between harmonics (the 10th percentile of the 0.5–8 kHz bins of a 4 096-point STFT in frames within 40 dB of the loudest, median over frames, against the original); stereo width, crest and peak columns. Means are over the seven music clips (Table 6).

5.2 RIPPLE AND SEAMS, CLIP BY CLIP

The ×3 decoders bring the 100 Hz ripple down to the original's level or near it: the duduk at +2.0 (v1) and +0.4 (v2) against Standard's +6.5, the orchestra at +0.8 / +1.0 against +7.2, Estas Tonne live at −0.2 / −0.8 against +3.5 (Figure 1 and Table 7). The 6 kHz line on Estas Tonne's studio recording falls from +20.6 to

TABLE 6 The headline, means over the seven music clips (Δ: decode minus original, dB). The judge is on clips 06, 07, 08. Speed: best of 24 decodes of one 30 s tile on an RTX 3090 shared with other jobs, float32. Bold: the best of the five in each row: the lowest judge, mel, valley Δ, decode time and memory, the highest SI-SDR, and the band, ripple and seam Δ closest to zero.

measure	Standard	Legacy	Mix 0.666	x3_v1	x3_v2
parameters, M	66.37	66.37	66.37	200.76	200.76
judge score (lower is better)	1.952	2.658	2.191	**1.320**	1.518
SI-SDR, dB (higher is closer)	**8.54**	8.43	8.53	7.85	7.88
mel distance (lower is closer)	**0.318**	0.395	0.340	0.339	0.340
Δ band 4–8 kHz	−0.82	−0.92	−0.92	**−0.08**	−0.25
Δ band 8–12 kHz	**−1.10**	−1.22	−1.22	−1.23	−1.24
Δ band 12–16 kHz	−2.53	−2.10	−2.93	**−1.03**	−1.14
Δ band 16–20 kHz	−3.19	−9.06	−6.43	**−1.58**	−1.64
100 Hz ripple Δ	+2.71	+3.58	+3.02	+0.60	**+0.36**
6 kHz line Δ	+3.59	+5.03	+3.69	+0.98	**+0.79**
12 kHz line Δ	+6.96	+9.74	+7.51	+2.16	**+2.13**
floor between harmonics Δ	+1.22	+1.68	**+1.10**	+1.59	+1.65
30 s decode, s	**0.29**	0.30	0.33	0.57	0.56
peak VRAM, GiB	**1.79**	**1.79**	**1.79**	2.30	2.30

+1.8 / +0.3 dB, on the reader from +8.8 to +1.3 / +2.8. The 12 kHz line leaves the reader (+0.3 / −0.7 against +12.9) and falls on most music clips, though not on all: on the duduk it is unchanged (+3.6 / +3.6 against Standard's +3.5, the original +4.2), and on both Estas Tonne recordings it now sits under the original's own line (studio +2.7 / +3.1 against the original's +6.0, live +2.3 / +2.0 against +4.1). On the shofar's decay it still stands at +13.4 / +13.0 dB, about 17 dB under Standard's +30.0 but there.

Three things go the other way at 6 kHz. On Estas Tonne's studio recording the ×3 line sits 3.7 and 5.2 dB under the original's own line, so real 6 kHz content of the recording goes with the seam; on the rock ballad it stands at +3.9 / +3.6 where Standard has 0.0; on the duduk at +2.0 / +2.7 where Standard has +0.6 and the original −2.4. Across the spectrum (Figure 7a), ×3 keeps more at 4–8 kHz and above 12 kHz; at 8–12 kHz all five lose about the same.

5.3 SIX GENERATED SONGS

Six songs of Ruach Studio's RS-TW fine-tune, decoded again from their own latents. There is no original: each song's audio is Standard's decode, so the seams and the ripple stand absolutely, and nothing else here is fidelity. Standard's ripple shows on two of them, +2.9 dB on the gospel song and +3.0 on Πάτερ ἡμῶν; ×3 brings them to +0.1 / −0.2 and +0.9 / +0.4 (Table 8). The 12 kHz line, up to +4.7 in Standard, stays at +0.5 to +2.4 in ×3: lower on four songs, higher on Abba (+0.2 → +0.9 / +1.2) and Πάτερ ἡμῶν (+0.8 → +2.0 / +2.0). At 6 kHz Standard shows no line on these songs (−0.4 to +1.0), and ×3 stands 0.7 dB higher on average, higher by more than 0.2 dB on four of the six. §5.4 shows what this measure hides.

5.4 THE RING: 40 GENERATED SONGS, AND THE LINE IN THE NOTCH

The Ring is Ruach Studio's fixed benchmark of generated songs: 20 cards (lyrics, style, language) × 2 seeds. Its side for RS-TW 2.0 at merge weight 0.75, 40 songs, was decoded again from each song's own latents by v1 and v2, each with the song's own level rule: a song's audio is the Standard decode scaled down by its

TABLE 7 Ripple and seams per clip, dB, absolute values with the original's own in grey beside each. Bold: per clip and measure, the decoder closest to the original's value.

	100 Hz ripple				6 kHz line				12 kHz line			
clip	orig.	Std	v1	v2	orig.	Std	v1	v2	orig.	Std	v1	v2
06 shofar	−1.5	−2.2	**−1.1**	−0.6	+1.2	+3.6	**+1.8**	+2.6	+0.6	+30.0	+13.4	**+13.0**
07 Estas Tonne, studio	+0.1	+1.6	**+1.0**	**+1.0**	+5.5	+20.6	**+1.8**	+0.3	+6.0	**+7.6**	+2.7	+3.1
08 Estas Tonne, live	0.0	+3.5	**−0.2**	−0.8	+0.6	+1.7	+1.3	**+1.1**	+4.1	+6.8	**+2.3**	+2.0
37 reader (speech, 24 kHz source)	+0.8	**+0.8**	+1.0	+1.0	+3.0	+8.8	+1.3	**+2.8**	−0.9	+12.9	+0.3	**−0.7**
05 duduk	+0.6	+6.5	+2.0	**+0.4**	−2.4	**+0.6**	+2.0	+2.7	+4.2	+3.5	**+3.6**	**+3.6**
12 orchestra (Shchedrin)	+0.2	+7.2	**+0.8**	+1.0	+0.7	+4.3	+2.8	**+1.8**	+0.8	+4.3	+2.8	**+2.1**
26 rock ballad (Scorpions)	−0.5	+0.7	**+0.1**	**+0.1**	+1.1	**0.0**	+3.9	+3.6	+0.9	+7.2	**+4.1**	+4.7
34 drum solo (Petra)	−0.6	−0.1	−0.1	**−0.2**	+1.3	+2.4	**+1.4**	**+1.4**	−0.6	+5.4	**+2.3**	+2.5

TABLE 8 Six generated songs: 100 Hz ripple and the two seams, dB, absolute (there is no original). Bold: the lowest of the three per measure; all equal, none. Decoded with the exact 60-frame halo.

		100 Hz ripple			6 kHz line			12 kHz line		
song	s	Std	v1	v2	Std	v1	v2	Std	v1	v2
Psalm 22 (Greek)	229.7	−0.2	−0.2	−0.2	**−0.4**	+1.0	+0.7	+4.7	**+1.9**	+2.4
Your Mercy (gospel)	213.8	+2.9	+0.1	**−0.2**	**0.0**	+1.6	+1.8	+3.1	+1.6	**+1.5**
Ring 25 (industrial metal)	185.6	−0.5	**−0.6**	−0.5	**+1.0**	**+1.0**	+1.1	+2.4	**+0.5**	+1.1
Abba (Hebrew, rock)	182.6	+0.3	0.0	**−0.2**	**+0.9**	+1.3	+1.4	**+0.2**	+0.9	+1.2
Πάτερ ἡμῶν (Greek)	85.8	+3.0	+0.9	**+0.4**	**+0.3**	+1.3	+1.1	**+0.8**	+2.0	+2.0
Ring 20 (Chinese folk fusion)	161.0	+0.4	−0.6	**−0.8**	+0.8	+0.8	**+0.7**	+2.8	**+1.1**	+1.2

peak when over full scale, which all 40 needed, and the Standard re-decode matches each song at 42.5 dB SI-SDR or better. The Ring's own measures were run unchanged on the song as it is, on v1 and on v2 (Table 9); Whisper large-v3 [20] gives the word error, re-heard on the GPU for all three sides, since the device alone moves the song's own error from 22.96 to 22.15 %.

Intelligibility is about the same: Whisper's error is 22.15 % on the song against 22.96 / 23.05 %, a mean rise of 0.81 / 0.90 points with a median change of 0, and fewer errors on 15 of the 40 songs. The decoder's own 25 Hz buzz is lower (−0.96 dB, better in 31 of 40), and stray tones fall from 2.15 to 0.80 per song on average. The sound is a little brighter, presence +0.43 dB and tilt +0.45 dB per octave, and the 6–16 kHz "sand" measure is slightly worse (+0.28 dB, better in only 10 of 40). The 100 Hz ripple is small on generated songs in Standard already (+0.02 dB) and stays small.

THE SEAMS, FINELY At 0.73 Hz resolution (Welch, 65 536-point segments) the 6 and 12 kHz seams of ×3 are two things at once: a sharp top (±1.5 Hz) that stands higher than Standard's, and around it a notch at ±3–15 Hz several dB deep (Figure 6). Read from the mean profiles over the 40 songs, at 6 kHz the top is +3.68 dB in the song against +5.46 / +5.36 (v1 / v2) and the notch −1.08 against −5.44 / −5.72; at 12 kHz the top is +4.58 against +6.48 / +6.82 and the notch −0.99 against −5.90 / −5.95.[2] An average over a few hertz around the line, as the clip measure of §2.2 (±6 Hz) and the judge's line6k (±9 Hz) take it, mixes the taller top and the deeper notch into one number that reads lower than Standard's: the Ring's ±9 Hz reading is −3.06 dB against +0.03 at 6 kHz. So the honest statement is that ×3 did not remove the seam; it narrowed

[2]The table's per-song means of the same quantities differ from the mean profiles by up to 0.2 dB.

TABLE 9 The Ring, 40 songs, means. The song as it is is Standard's decode. The paired columns give the mean change against the song and, for measures with a direction, in how many of the 40 songs the decoder did better. Bold: the best of the three where a measure has a direction (a line's top: lower is better).

measure	the song (Standard)	x3_v1	x3_v2	v1 − song · better in	v2 − song · better in
Whisper word error, %	**22.15**	22.96	23.05	+0.81 · 15 of 40	+0.90 · 15 of 40
decoder 25 Hz buzz, dB	11.57	10.61	**10.50**	−0.96 · 31 of 40	−1.07 · 37 of 40
stray tones, count	2.15	0.80	**0.62**	−1.35 · 21 of 40	−1.52 · 22 of 40
clicks, count	34.45	**32.75**	33.92	−1.70 · 15 of 40	−0.53 · 8 of 40
noise floor, dBFS	**−27.14**	−27.00	−27.07	+0.14 · 21 of 40	+0.07 · 19 of 40
sand 6–16 kHz, dB	**−17.89**	−17.61	−17.59	+0.28 · 10 of 40	+0.30 · 10 of 40
top, kHz	**15.23**	15.02	15.00	−0.21 · 13 of 40	−0.24 · 13 of 40
presence 2–5 kHz, dB	−17.31	−16.88	−16.95	+0.43	+0.37
spectral tilt, dB per octave	−5.20	−4.75	−4.76	+0.45	+0.44
crest, dB	16.17	16.07	16.13	−0.10	−0.04
loudness, LUFS [21]	−14.61	−14.65	−14.72	−0.04	−0.12
true peak, dBTP	−0.27	**−0.41**	**−0.41**	−0.14 · 19 of 40	−0.14 · 21 of 40
100 Hz ripple, dB	0.02	−0.07	**−0.13**	−0.10 · 24 of 40	−0.15 · 23 of 40
6 kHz line top, dB	**3.70**	5.46	5.36	+1.76 · 11 of 40	+1.66 · 14 of 40
6 kHz skirt (±3–15 Hz), dB	−1.31	−5.51	−5.81	−4.20	−4.50
12 kHz line top, dB	**4.58**	6.48	6.82	+1.90 · 7 of 40	+2.24 · 6 of 40
12 kHz skirt, dB	−0.98	−5.84	−5.95	−4.86	−4.97
18 kHz line top, dB	11.13	**5.45**	5.66	−5.67 · 34 of 40	−5.47 · 34 of 40

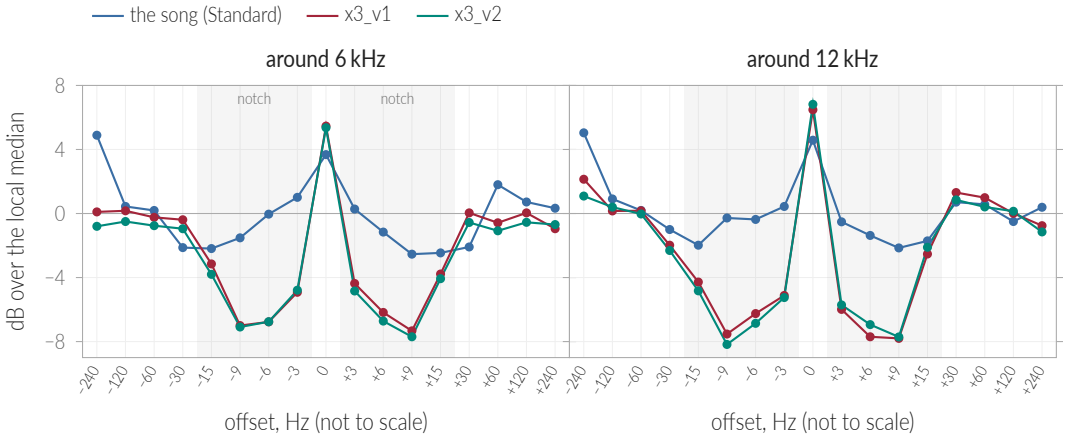


FIGURE 6 Mean spectrum around 6 and 12 kHz over the 40 Ring songs, 0.73 Hz bins, dB over the median of 300–1 000 Hz around the line; the offsets are evenly spaced, not to scale. Shaded: the ±3–15 Hz skirt. The ×3 decoders: a taller centre and a notch on both sides.

the line and dug a notch around it. At 18 kHz, three times 6 kHz, the picture is the reverse: Standard's top stands at +11.12 dB, ×3's at +5.45.

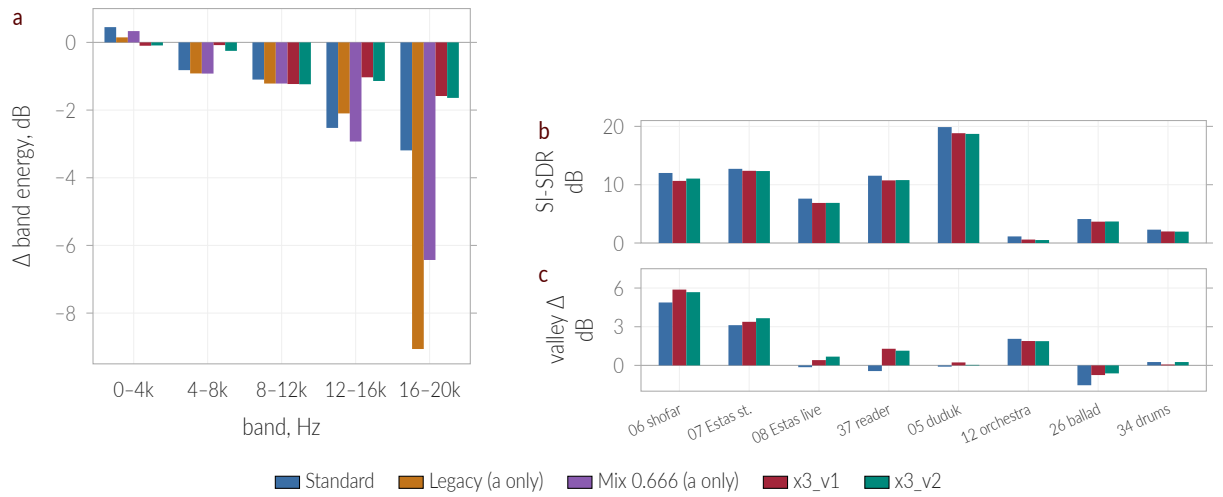


FIGURE 7 (a) Band energy against the original, mean of the seven music clips, all five decoders: ×3 keeps more at 4–8 kHz and above 12 kHz, and at 8–12 kHz all five lose about the same. (b) SI-SDR against the original per clip (higher is closer): lower for ×3 on every clip. (c) The floor between harmonics, Δ against the original per clip (lower is better). In (b) and (c): Standard, x3_v1, x3_v2.

6 WEAKNESSES AND COSTS

THE FLOOR BETWEEN HARMONICS IS NOT BETTER On the shofar's decay the first two thirds stand at +6.38 / +5.38 dB (v1) and +6.04 / +5.13 (v2) over the original, against Standard's +5.49 / +5.95. Over the music the valley measure averages +1.59 / +1.65 dB against Standard's +1.22: the noise between harmonics is 0.4 dB higher on average, higher on five of the seven music clips and lower or within 0.1 dB on the rest (Figure 7c). 150 000 steps with the valley term at weight 2 did not move it; it may be capacity rather than emphasis.

WAVEFORM FIDELITY IS LOWER SI-SDR averages 7.85 / 7.88 dB against 8.54, lower on every clip by 0.3–1.4 dB (Figure 7b). The mel distance averages 0.339 / 0.340 against 0.318, about Mix 0.666's (0.340). The ×3 decoders were trained by spectral and adversarial terms, not by the waveform, and they place phase differently from the original; whether that is heard is for the listener's ear, not for this number.

8–12 KHZ All five decoders lose 1.10–1.24 dB there on average, and on the shofar and Estas Tonne's studio recording the ×3 decoders lose 3.1 and 3.0 dB (v1; v2 2.9 and 3.1) where Standard loses 0.6 and 1.6. This is the thinnest place of ×3.

ABOVE THE READER'S BAND The reader's original is silent above 12 kHz; there the ×3 decoders make about 4 dB more floor of their own (+30.6 against +26.7 dB over that silence at 16–20 kHz, some 85–90 dB under the voice), which raises its multi-STFT distance by about half (1.48 against 1.01).

PEAKS On every music clip but the shofar the ×3 decoders overshoot less and undershoot more than Standard (peak column p90 +0.26 to +1.18 against Standard's +0.56 to +1.89), so their peaks sit a little low.

SPEED, MEMORY AND TILING About twice the decode time and three times the weights (Table 10). Peak memory stays flat with length because of tiling, but the tiles need 60 latent frames (2.4 s) of context on each side instead of 12, which costs about 11 % more arithmetic at 750-frame cores than a 16-frame halo did.

TABLE 10 Speed and memory, float32, RTX 3090 shared with other jobs: a fresh process per decoder, three interleaved rounds of eight decodes after a warm-up; best is the decoder's own speed, median what the shared card gave. Peak VRAM is the decode's own allocation, weights included. The 180 s figures used the 16-frame halo; the exact one adds about 11 % arithmetic. Bold: the best in each column.

decoder	params, M	weights, GiB	30 s: best / median, s	× real time	30 s VRAM, GiB	180 s: best / median, s	180 s VRAM, GiB
Standard	66.37	**0.25**	**0.29** / 0.41	**103**	**1.79**	1.93 / 2.63	**1.91**
Legacy	66.37	**0.25**	0.30 / **0.35**	100	**1.79**	**1.88** / **2.25**	**1.91**
Mix 0.666	66.37	**0.25**	0.33 / 0.40	91	**1.79**	**1.88** / 2.60	**1.91**
x3_v1	200.76	0.75	0.57 / 0.62	52	2.30	3.71 / 5.14	2.41
x3_v2	200.76	0.75	0.56 / 0.59	53	2.30	4.06 / 5.25	2.41

TABLE 11 Error of half-precision decodes against the float32 decode, on 30 s of a song: energy of the error under the music's own, per band, dB.

	whole	0–4 kHz	4–8 kHz	8–12 kHz	12–16 kHz	16–20 kHz
x3_v1 in bf16	−37.2	−38.1	−24.0	−25.4	−24.0	−22.8
x3_v1 in fp16	−54.8	−56.2	−39.8	−42.2	−39.9	−38.3
Standard in bf16	−38.1	−39.4	−24.0	−23.9	−22.3	−22.2
Standard in fp16	−56.4	−57.6	−42.3	−41.6	−40.0	−38.7

7 NUMERICAL IDENTITY ACROSS ENGINES

7.1 FLOAT32 ONLY

The decoder was trained, judged and measured in float32 with TF32 off, and the release is float32 only: there is no bf16, fp16 or quantised copy, and the integration kit's loader refuses one. Half precision puts its error where this decoder's gains are (Table 11): above 4 kHz the error of bf16, 22–25 dB under the music, is as large as the band differences between Standard and ×3 themselves; fp16, at 38–40 dB, is ten times closer and still not the decoder that was measured. TF32 matters too: PyTorch lets cuDNN convolutions run in TF32 by default on Ampere and newer cards, and left on, a decode misses the verification tolerance of 10^{-4} tenfold ($1.1 \cdot 10^{-3}$ measured). Stock ComfyUI [24] picks fp16 for the YuE2 VAE on most NVIDIA cards.

7.2 THE SAME DECODER IN A GGML ENGINE

The same holds in Ruach Studio's ggml [23] engine, which reads GGUF copies of the two decoders: the weights and the convolutions' unfolded inputs stay float32. On CUDA, ggml lets cuBLAS run float32 products in TF32 on every handle, which left an RTX 3090's decode only 55–67 dB from the reference; the engine now asks for true float32 on this decoder's products, through a ggml patch that turns TF32 off for those calls only, while Standard decodes bit for bit as before. Against the kit's reference (Table 12) the RTX 3090 stands −111 to −115 dB (v1) and −113 to −116 dB (v2) away, an Intel Arc Pro B50 −114 to −118 dB, the CPU −108 to −116 dB. A 59.2 s song decoded by the engine on the Arc B50 in 512-frame tiles with the 60-frame halo stands −118.5 dB from its whole decode at the tile seams: the seams are not there. That song took 6.2–11.4 s on the Arc (2.96 GB of working buffer, 0.77 GB of weights) and 1.36 s on an RTX 3090.

7.3 TILING: THE 60-FRAME HALO

Each tile needs 60 latent frames (2.4 s) of context on each side to equal the whole decode; Standard needs 12. With a 16-frame halo, a 214-second song differs from its whole decode by up to 0.16 in amplitude near a

TABLE 12 The decode of the kit's reference latents (64 s, four tiles of 512 frames in the engine) against the reference made by PyTorch on CUDA in float32 with TF32 off: error energy under the signal in four 4 096-sample excerpts (the start, both tile seams, inside the last tile), dB. Standard's engine path is the older F16 one, kept bit for bit (equal PCM checksums before and after).

	x3_v1				x3_v2				
engine, backend	start	seam 1	seam 2	last tile	start	seam 1	seam 2	last tile	64 s decode
ggml, RTX 3090, TF32 on (before)	−63.2	−64.4	−65.5	−55.4	−63.9	−64.5	−66.7	−65.7	
ggml, RTX 3090, true float32	−110.9	−113.4	−114.5	−112.5	−112.9	−113.5	−115.8	−113.3	2.1 s
ggml, Intel Arc Pro B50 (SYCL)	−113.8	−116.2	−118.2	−116.1					13.8 s
ggml, CPU				−108 to −116					
PyTorch kit, CPU				max |diff| $1.6 \cdot 10^{-6}$: pass					85 s

30-second tile seam (−27 to −51 dB under the music in the second around each seam); with the exact halo the tiled decode equals the whole decode to $8 \cdot 10^{-6}$. The integration kit's tiled decode uses the exact halo by default and refuses a shorter one, and reports it under the same name and meaning as m-a-p's own `required_halo()`, so a halo of max(16, `required_halo(core)`) is right for both decoders. Every whole-song number in this report was measured on exact decodes.

7.4 ONE TRAP, MEASURED

Unpatched ComfyUI does not load this decoder, and it loses it without a word. It recognises the YuE2 VAE inside a checkpoint by its key names, builds the decoder with Standard's shape and loads it non-strictly: the 210 tensors of the new units are dropped without an error, and the result is neither decoder (23 % of the signal's RMS away from the real ×3, peaks off by 0.54). The integration guide [25] gives two ways that load it whole, a node pack and a patch for ComfyUI that decodes bit for bit like the reference. The kit's verifier with its reference file (1 600 seeded latent frames, 64 s, three tiles, five excerpts of v1, v2 and Standard made on CUDA in float32 with TF32 off and the exact halo) is the proof of any integration: v1, v2 and Standard on CUDA 0.0; v1 on the CPU $1.6 \cdot 10^{-6}$; v1 checked against v2's reference 0.19, a failure as it should be. The tolerance is 10^{-4}.

8 RELEASE AND LICENCES

Each release is one float32 `.safetensors` file, decoder only: 427 tensors, 803 071 224 bytes, with its configuration, variant, step and description in the metadata. Beside each stands the same decoder as the ggml engine reads it, a float32 GGUF of 803 056 768 bytes with the shape and the 60-frame halo in its keys (Table 13). They are in `VAE-RS/` of the repository `goldhub/YuE2-RS-Finetunes` on Hugging Face [25], with their checksums, the weight licence and the integration kit; the study behind this report, with every figure, hour and loss, is in `docs/` of the same repository. Encode with m-a-p's Standard encoder as always: the files have no encoder. Ruach Studio [26] 2.0.7 fetches v1 with every install and decodes with it by default.

WEIGHTS CC BY-NC 4.0, the licence of m-a-p's YuE2-3B and YuE2-Vae from which they are adapted, and never stricter than our own fine-tunes beside them. Non-commercial use; credit YuE2 and m-a-p (model name and source repository, as their licence asks) and Ruach Studio for VAE-RS.

CODE The integration code (the decoder module, the verifier, the node pack, the patches) is under Apache-2.0, with a NOTICE that requires credit to Ruach Studio and Viktor Zhuromskyy in any redistribution

TABLE 13 The released files and their SHA-256.

file	SHA-256
YuE2-VAE-RS-x3_v1.safetensors	a88c8da0d0913115953196b01aaf770ea128ab224bde090a8f34ee46db3bb039
YuE2-VAE-RS-x3_v2.safetensors	114da484c564a265a694f019d0715cde2f87da88815a054f72ce77c5250f1a0c
YuE2-VAE-RS-x3_v1-F32.gguf	368a91278f0b25e67bc14b1a894f9e8c341c5fccb70afa880fbc1de6afdb4959
YuE2-VAE-RS-x3_v2-F32.gguf	93de6e8a18246670f4cf49d5f4f193aece95dda48463eda13a303e7f0d313f97

or derivative work (Apache-2.0 §4 d). A patch submitted to ComfyUI itself would be contributed under ComfyUI's GPL-3.0. The SnakeBeta activation and the Oobleck layout follow stable-audio-tools and BigVGAN, both MIT; their notices, which m-a-p ship with YuE2-Vae, go with the code. Training audio stays under its own sources' terms; the weights do not contain or reproduce it.

9 CONCLUSION

A decoder grown from Standard by units that start silent, trained for about 36 hours on one RTX 3090, brings the 100 Hz ripple of the YuE2 VAE to the originals' level on sustained material and lowers the 6 and 12 kHz seams on recorded music, while keeping more of the top octave. It does not remove the seams on generated songs; it narrows them into a taller line inside a deeper notch. It is further from the waveform than Standard, its floor between harmonics is a little higher, and it costs twice the time and three times the weights. Both runs drifted after their best day while the reconstruction kept improving, together with a widening ratio of the discriminators' learning rate to the generator's; the ratio is a suspect, not a finding. The release is float32 only, and the same decoder decodes identically in PyTorch and in a ggml engine on three kinds of hardware.

These numbers say where to listen: sustained dense tones such as strings, a duduk or a held chord for the fine 100 Hz crackle; cymbals, breath and the air for the seams; the low end of a kick for the sub-bass; a loud transient for the peaks. Decode the same latents with Standard and with x3_v1 or x3_v2, match the loudness, and judge by your own ear on your own music.

CONTRIBUTIONS AND ACKNOWLEDGEMENTS

YuE2 and its VAE, Standard and Legacy, are the work of m-a-p (Multimodal Art Projection); every number here begins from their decoder and their open weights. Viktor Zhuromskyy (ЙирмиЙáҐу), Ruach Studio: the decision to grow the decoder, the choice of the two releases and the judgement of every step. Claude (Anthropic): the code, the measurements, the study and this report, made with Viktor in what he calls the HERETICAL TANDEM. Training audio comes from the sources of §3.2, as packed in Ruach Studio's RS-TW dataset.

DATA AND PROVENANCE Every number in this report is read from the study's data files of 9 November 2026: the five decoders on the eight clips and the six songs, their speed and memory, the hours, steps, evaluations and losses of both runs, the Ring through v1 and v2, and the shared-pool test; the engine figures are from the engine's logs of 10 November 2026. The study [25] lists each file with the script that made it.

REFERENCES

[1] R. Yuan, J. Pan, J. Jiang, Z. Wu, Z. Zhou, J. Sun, et al. *YuE2: Unifying Symbolic and Audio Music Generation at Frontier Quality.* arXiv:2609.33757, 2026.

[2] Multimodal Art Projection (m-a-p). *YuE2-Vae* and *YuE2-Vae-legacy*, model cards and weights. `https://huggingface.co/m-a-p/YuE2-Vae`, 2026.

[3] R. Yuan, H. Lin, et al. *YuE: Scaling Open Foundation Models for Long-Form Music Generation.* arXiv:2503.08638, 2025.

[4] Y. Li, R. Yuan, G. Zhang, Y. Ma, X. Chen, H. Yin, et al. *MERT: Acoustic Music Understanding Model with Large-Scale Self-supervised Training.* ICLR 2024; arXiv:2306.00107.

[5] Z. Evans, J. D. Parker, C. J. Carr, Z. Zukowski, J. Taylor, J. Pons. *Stable Audio Open.* ICASSP 2025; arXiv:2407.14358.

[6] Stability AI. *stable-audio-tools* (the Oobleck autoencoder). `https://github.com/Stability-AI/stable-audio-tools`.

[7] S. Lee, W. Ping, B. Ginsburg, B. Catanzaro, S. Yoon. *BigVGAN: A Universal Neural Vocoder with Large-Scale Training.* ICLR 2023; arXiv:2206.04658.

[8] L. Ziyin, T. Hartwig, M. Ueda. *Neural Networks Fail to Learn Periodic Functions and How to Fix It.* NeurIPS 2020; arXiv:2006.08195.

[9] A. Odena, V. Dumoulin, C. Olah. *Deconvolution and Checkerboard Artifacts.* Distill, 2016. doi:10.23915/distill.00003.

[10] J. Pons, S. Pascual, G. Cengarle, J. Serrà. *Upsampling Artifacts in Neural Audio Synthesis.* ICASSP 2021; arXiv:2010.14356.

[11] T. Salimans, D. P. Kingma. *Weight Normalization: A Simple Reparameterization to Accelerate Training of Deep Neural Networks.* NeurIPS 2016; arXiv:1602.07868.

[12] J. Kong, J. Kim, J. Bae. *HiFi-GAN: Generative Adversarial Networks for Efficient and High Fidelity Speech Synthesis.* NeurIPS 2020; arXiv:2010.05646.

[13] A. Défossez, J. Copet, G. Synnaeve, Y. Adi. *High Fidelity Neural Audio Compression.* TMLR 2023; arXiv:2210.13438.

[14] X. Mao, Q. Li, H. Xie, R. Y. K. Lau, Z. Wang, S. P. Smolley. *Least Squares Generative Adversarial Networks.* ICCV 2017; arXiv:1611.04076.

[15] K. Kumar, R. Kumar, T. de Boissiere, L. Gestin, W. Z. Teoh, J. Sotelo, A. de Brébisson, Y. Bengio, A. Courville. *MelGAN: Generative Adversarial Networks for Conditional Waveform Synthesis.* NeurIPS 2019; arXiv:1910.06711.

[16] M. Defferrard, K. Benzi, P. Vandergheynst, X. Bresson. *FMA: A Dataset for Music Analysis.* ISMIR 2017; arXiv:1612.01840.

[17] D. Bogdanov, M. Won, P. Tovstogan, A. Porter, X. Serra. *The MTG-Jamendo Dataset for Automatic Music Tagging.* ICML 2019 Workshop on Machine Learning for Music Discovery.

[18] P. D. Welch. *The Use of Fast Fourier Transform for the Estimation of Power Spectra.* IEEE Transactions on Audio and Electroacoustics 15(2):70–73, 1967.

[19] J. Le Roux, S. Wisdom, H. Erdogan, J. R. Hershey. *SDR – Half-baked or Well Done?* ICASSP 2019; arXiv:1811.02508.

[20] A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey, I. Sutskever. *Robust Speech Recognition via Large-Scale Weak Supervision.* ICML 2023; arXiv:2212.04356.

[21] ITU-R. *Recommendation BS.1770: Algorithms to Measure Audio Programme Loudness and True-Peak Audio Level.*

[22] A. Paszke, S. Gross, F. Massa, A. Lerer, et al. *PyTorch: An Imperative Style, High-Performance Deep Learning Library.* NeurIPS 2019; arXiv:1912.01703.

[23] G. Gerganov and contributors. *ggml: Tensor Library for Machine Learning.* `https://github.com/ggml-org/ggml`.

[24] *ComfyUI.* `https://github.com/comfyanonymous/ComfyUI`.

[25] Ruach Studio. *YuE2-RS-Finetunes*: `VAE-RS/` (weights, kit), `docs/VAE-RS-x3-study.html` (the study), `docs/VAE-RS-x3-integration.md` (the integration guide). `https://huggingface.co/goldhub/YuE2-RS-Finetunes`, 2026.

[26] Ruach Studio. `https://github.com/igrbible/Ruach_Studio`.